

Srinivasan Poonkundran

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PROFESSIONAL EXPERIENCE

Research Assistant

Feb. 2026 – Present

The University of Arizona

Tucson, AZ

- Built and evaluated machine learning pipelines for large-scale sensor and behavioral datasets using Python and scikit-learn.
- Implemented anomaly detection workflows using Isolation Forest, Local Outlier Factor (LOF), and deep autoencoder-based models for time-series analysis.
- Developed reproducible preprocessing, evaluation, and experimentation pipelines for data-driven research workflows.
- Worked with large-scale time-series datasets containing over 1M+ records for behavioral analysis and anomaly detection studies.
- **Skills:** Python, AWS, SQL, Autoencoder, Transformers, TensorFlow, PyTorch

Programmer Analyst — Full-Stack Software Engineer

Sept. 2021 – Jan. 2024

Cognizant Technology Solutions

Remote

- Built Java Spring Boot microservices deployed on AWS, backed by MySQL databases, secured with OAuth2-based authentication and authorization, and integrated with Angular dashboards used by internal stakeholders.
- Improved API response time by 35% by optimizing SQL queries and caching critical endpoints.
- Built a Python-based PDF label validation tool to automate manual checks, reducing operational effort and saving approximately \$1.6K annually.
- Collaborated with cross-functional Agile teams to incorporate feedback into feature development and iteration.
- Contributed to code reviews and refactored backend services with Java and Spring Boot, improving code quality and reducing defect occurrences, which accelerated release cycles.
- Mentored 3 junior developers through weekly code reviews and pair programming sessions, reducing the number of code review defects per sprint from 25 to 20.
- **Skills:** Java, Spring Boot, Unit Testing, Angular, Python, AWS, MySQL, Microservices, CI/CD, Performance Optimization, Problem Solving, Data Analytics & Visualization, Docker, Agile, Cross-functional Collaboration

Software Developer

Jan. 2023 – Jan. 2024

Self-Employed

Part-time, Remote

- Developed a full-featured Android application to manage invoices, sales, inventory, and staff operations for a retail clothing store handling more than 1,000 monthly transactions.
- Designed and deployed interactive sales dashboards with visual charts to track daily revenue, top-selling items, seasonal demand trends, and customer purchase history, improving business decision-making for 3 store managers.
- Integrated WhatsApp API for automated invoice sharing, enabling instant digital receipts to 700+ recurring customers and reducing paper invoice costs by \$500 annually.
- Configured AWS S3 cloud storage to securely back up over 12,000 invoices and sales records, providing cross-device access and reducing risk of local data loss by 100%.
- Collaborated with the client remotely through weekly demo sessions, collected feedback, and delivered 4 incremental releases over 12 months using Agile methodology.
- **Skills:** Android, Java, AWS S3, SQL, Data Visualization, WhatsApp API, UI/UX Design

INTERNSHIP EXPERIENCE

Xerxez

Oct. 2020 — Dec. 2020

Cloud Computing & Machine Learning Intern

Remote

- Completed a 6-week internship focused on Cloud Computing and Machine Learning using Amazon Web Services (AWS), gaining hands-on experience with cloud infrastructure and project deployment.
- Assisted in developing machine learning models and deploying them on AWS, enhancing understanding of cloud-based data workflows.
- Collaborated with the Project Division team on AWS-based ML projects, learning best practices for cloud project management and data processing pipelines.
- Demonstrated initiative and curiosity by exploring advanced ML concepts and AWS services beyond the assigned tasks.
- **Skills:** AWS, Cloud Computing, Machine Learning, Python, Data Analysis, Project Collaboration

LEADERSHIP EXPERIENCE

Acadflip – EdTech Startup (“The Book of Video Lectures”)

Jan. 2020 – Sept. 2020

Brand Ambassador

Remote

- Appointed as an official Brand Ambassador of Acadflip, representing the platform among students and educators.
- Promoted Acadflip’s video lecture platform on campus and social media, reaching an audience of nearly 500 peers.
- Organized awareness campaigns and peer learning sessions to drive student engagement and increase platform adoption.
- Acted as a liaison between students and the startup team, providing feedback that helped improve product features and user experience.
- **Skills:** Campus Outreach, Public Speaking, Social Media Promotion, Community Engagement

TECHNICAL SKILLS

Languages: Python, R, Java, C++, C, JavaScript, TypeScript, SQL

Data Science & Machine Learning: NumPy, pandas, scikit-learn, XGBoost, Matplotlib, seaborn, Keras, TensorFlow, StatsModels

Development Practices: Object-Oriented Programming, Functional Programming, Microservice Architecture, MVC Pattern, Agile, Test-Driven Development, CI/CD

Databases & Querying: MySQL, PostgreSQL, MongoDB, SQL/NoSQL

Cloud & DevOps: AWS, Docker

Web & Backend Technologies: Angular, Spring Boot, Spring MVC, Node.js, Express.js, REST APIs, JWT, Maven, HTML, CSS

Tools & IDEs: Jupyter Notebook, PyCharm, VS Code, Android Studio, RStudio, IntelliJ, Git, GitHub, Postman, Docker Hub

Visualization & Reporting: Power BI, Tableau, Matplotlib, Seaborn, Plotly

Certifications: Amazon Web Services Certified Cloud Practitioner | [Credentials](#)

PROJECTS

GradeLens - RAG Powered Auto Grader | [GitHub](#)

- Architected a production-grade Retrieval-Augmented Generation (RAG) system using vector embeddings and LLM orchestration to enable scalable, context-aware automated grading.
- Designed modular ML pipelines for embedding generation, semantic retrieval, rubric validation, and structured output enforcement.
- Implemented asynchronous API handling with parallel execution to reduce grading latency across multiple exam questions.
- Deployed the system within a Django + React application, enabling scalable, low-latency inference.
- **Skills:** Retrieval Augmented Generation (RAG), Hugging Face, Natural Language Processing (NLP), Large Language Models (LLMs), Python, Django + Django REST framework, React, Document Ingestion

Emotion Detection from Text | [GitHub](#)

- Designed and trained a transformer-based NLP classification system using BERT embeddings and Bi-GRU layers, achieving 0.83 F1-micro across 7 emotion classes.
- Built scalable preprocessing and training pipelines using Hugging Face and TensorFlow to efficiently handle large text datasets.
- Optimized model performance through structured tuning and regularization to improve generalization.
- **Skills:** BERT, Hugging Face, TensorFlow, Keras, GRU, Natural Language Processing (NLP), Python

Heart Rate Monitoring System | [GitHub](#)

- Developed an IoT-enabled health monitoring application to track heart rate and blood oxygen (SpO2) levels in real time using a MAX30102 sensor and Particle Photon board.
- Built secure RESTful APIs with Node.js, Express, and MongoDB for reliable sensor data capture, retrieval, and user-specific storage.
- Created a responsive React based frontend with interactive daily and weekly charts to visualize real-time physiological data.
- Implemented multi-user support, configurable data collection intervals, and offline storage with automatic synchronization after reconnecting.
- **Skills:** Internet of Things (IoT), Node.js, Express, MongoDB, React, REST APIs, Data Visualization, Real-Time Charts

Cricket Match Result Forecasting | [GitHub](#) | [Website](#)

- Developed a Random Forest classifier to predict ODI cricket match outcomes, achieving 87% accuracy on historical data.
- Performed data preprocessing and feature engineering using player form, team performance, and venue-specific factors to improve prediction accuracy.
- Built backend APIs to serve match statistics and dynamically updated predictions using simulated live match data.
- Created an interactive dashboard that refreshed predictions every five minutes and visualized changing match outcomes.
- **Skills:** Random Forest Regressor, Python, Java, MySQL, API Design, Cricket Analytics, Real-Time Prediction, Data Preprocessing, Feature Engineering

Cricket Metrics | [GitHub](#) | [Website](#)

- Developed a cricket analytics dashboard using over 150,000 historical data points to analyze player, team, and match performance.
- Built backend APIs for batting, bowling, & partnership data, supporting simulated live match updates and analysis.
- Created a Quarto-based interactive dashboard with line, bar, donut, and map visualizations for scoring patterns and match trends.
- Implemented efficient data processing pipelines to handle large datasets while maintaining responsive and accurate dashboard performance.
- **Skills:** R, Shiny, Quarto, Java, MySQL, API Development, Data Visualization, ggplot2, Cricket Analytics, Real-Time Dashboards, Data Processing

Psychometric Analysis | [GitHub](#) | [Website](#)

- Conducted a psychometric analysis on 890 fictional characters across 100 universes using data from the Open-Source Psychometrics Project.
- Performed extensive data cleaning, feature engineering, and correlation analysis to uncover meaningful links between traits, audience engagement, and character prominence.
- Visualized insights with bar charts, density plots, and radar charts to highlight trends in personality traits and audience response.
- Developed interactive dashboards to allow exploration of relationships between MBTI types, psychometric scores, and character popularity.
- **Skills:** R, Quarto, Data Analysis, Data Visualization, ggplot2

COVID-19 Protocol Monitoring System | [GitHub](#) | [Published in IJRASET](#)

- Developed a COVID-19 surveillance system combining mask detection, social distancing monitoring, temperature sensing, and automated sanitization.
- Implemented real-time mask and social distancing detection using MobileNetV2, OpenCV, and YOLO with the COCO dataset.
- Designed a client-server architecture using Raspberry Pi Pico and Arduino Uno nodes to support monitoring across multiple entry and exit points.
- Built a Flask-based web interface for live surveillance feeds and reports, and published the research in IJRASET.
- **Skills:** Computer Vision, CNN, MobileNetV2, YOLO, Socket Programming, Raspberry Pi, Arduino, Flask, OpenCV, SQL, IoT

FBI Crime Data Reporting and NIBRS Adoption Rate Analysis | [GitHub](#)

- Analyzed 19,000+ U.S. law enforcement agency records from the FBI Uniform Crime Reporting Program to examine agency distribution and NIBRS participation.
- Preprocessed the dataset by imputing missing coordinates, classifying agency types, and standardizing participation dates for geographic and temporal analysis.
- Created visualizations across all 50 states to compare agency structures and track NIBRS adoption trends over time.
- Identified urban-rural differences and uneven participation patterns, highlighting states and agencies lagging in standardized crime reporting.
- **Skills:** Python, Pandas, Data Cleaning, Exploratory Data Analysis (EDA), Data Visualization, Matplotlib, Seaborn

International Patients Treatment Management System | [GitHub](#)

- Developed a full-stack healthcare management system for international patient registration, treatment tracking, and cross-border case coordination.
- Built Spring Boot microservices with Feign Client and secure RESTful APIs for patient records, medical workflows, and billing operations.
- Created a responsive Angular dashboard with role-based interfaces for doctors, administrators, and patients.

- Implemented multi-currency billing, authentication, authorization, and RBAC to support secure international healthcare transactions.
- **Skills:** Java, Angular, Spring Boot, Spring MVC, Microservices, FeignClient, REST APIs, H2 Database

Flight Fare Prediction | [GitHub](#)

- Built a Flask web application to predict flight fares using travel details such as airline, route, stops, journey date, and duration.
- Developed and integrated a Random Forest regression model that achieved 88% prediction accuracy on historical flight data.
- Implemented REST endpoints for input validation, preprocessing, real-time model inference, and dynamic fare prediction.
- Performed feature engineering, including datetime parsing, categorical encoding, and duration transformation, and created an HTML/CSS interface for user input.
- **Skills:** Flask, Python, Scikit-learn, Pandas, Pickle, Web Development

EDUCATION

The University of Arizona

Master of Science, Data Science

Honors: Distinguished Graduate Scholar (Fall 2025)

Tucson, AZ

Jan. 2024 – Dec. 2025

Savitribai Phule Pune University

Bachelor of Engineering, Computer

Pune, India

Aug. 2017 – May 2021

COURSEWORK

Core Data Science: INFO 511 – Foundations of Data Science, INFO 523 – Data Mining and Discovery, INFO 502 – Data Ethics

Machine Learning and NLP: INFO 557 – Neural Networks, INFO 539 – Statistical Natural Language Processing, INFO 555 – Applied Natural Language Processing

Data Engineering and Visualization: INFO 526 – Data Analysis and Visualization, INFO 531 – Data Warehousing in the Cloud

Web Development: ECE 513 – Web Development and the Internet of Things